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TUTORIAL

A Primer on Quantum Neural Network Model for Text Classification: From Concept to Design

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ABSTRACT Quantum neural networks (QNNs) have recently attracted remarkable attention as a potential alternative to classical and deep neural network models, particularly under constraints imposed by noisy intermediate-scale quantum (NISQ) devices that support only shallow circuits with a few quantum bits (qubits) and limited variational depth. However, applying QNNs to natural language processing remains challenging because text representations are typically high-dimensional, while NISQ classifiers can only encode a small number of features at a time, requiring classical compression to enable stable training and flexible circuit configuration. In addition, many existing studies integrate QNNs into rich-text representation pipelines, where the quantum circuit is often implemented as a variational quantum circuit (VQC). Here, the VQC acts as an additional component within a larger model, and it remains unclear how the quantum component improves the classification task. Therefore, in this work, we propose a fully reproducible and compact QNN pipeline that provides a controlled setting for studying the contribution of the VQC to text classification. The model is trained end-to-end with a lightweight classical compression layer, and the overall study is supported by systematic ablation analyses of the compression layer, the classifier heads under a shared frozen low-dimensional representation, and quantum circuit structure. Our model uses the term frequency-inverse document frequency features with linear compression followed by a shallow four-qubit, two-layer VQC, implemented in PyTorch and PennyLane for end-to-end training. It is evaluated on five benchmark text classification datasets against classical baselines, including logistic regression and multilayer perceptron models. Experimental results show that the proposed QNN achieves a competitive performance across all evaluated datasets, yielding the best results on three of the five benchmark datasets.

INDEX TERMS Quantum machine learning, quantum neural network, text classification, natural language processing.

I. INTRODUCTION

The evolution of computational paradigms from classical computing (CC) to quantum computing (QC) has significantly reshaped the formulation and solution of complex problems. In CC, information is represented using bits that take deterministic values of 0 or 1, while computations are carried out through logical operations implemented by classical logic gates such as AND, OR, and NOT. This

representation of classical information has supported decades of innovation, and continues to form the foundation of modern computing systems for a wide range of applications.

Quantum computing moves beyond this classical information representation by introducing quantum bits (qubits), which are represented in coherent superposition of $|0\rangle$ and $|1\rangle$ states and form entangled correlations that allow multiple computational states to be processed simultaneously. Rather than representing a single deterministic value, a qubit is described by a quantum state that assigns some probability amplitudes to the basis states. Unlike classical bits,

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a qubit encodes information probabilistically, with its state representing a weighted combination of possible outcomes. As multiple qubits interact, their information is jointly represented by a composite quantum state. These quantum states evolve under unitary transformations implemented by quantum gates such as the Hadamard (H) gate, Pauli gates (X , Y , Z), parameterized rotation gates such as $R_X(\theta)$, $R_Y(\theta)$, and $R_Z(\theta)$, controlled-NOT (CNOT) operations, and Ising-type interactions [1], enabling quantum systems to perform parallel computations within exponentially large state spaces and explore multiple computational pathways through superposition and entanglement.

Alongside these developments in QC, machine learning has become a cornerstone of modern data-driven intelligence, driving advances in computer vision, speech recognition, and natural language processing (NLP). Particularly, neural networks (NNs) demonstrate a strong capability in learning the hierarchical representations from large datasets, yet their effectiveness often depends on a large number of trainable parameters and substantial computational resources [2]. As the demand for efficient learning from increasingly high-dimensional data grows, quantum machine learning (QML) has emerged as a promising direction that explores the integration of quantum computation into the learning pipeline. Quantum neural networks (QNNs) form one part of the developed QML methods, where classical input data are encoded into quantum states and then processed by quantum gates with trainable parameters [3]. Quantum neural networks have been applied for both discriminative and generative learning tasks, the former being one of the most active research areas in QML, with applications including image classification, fraud detection, disease diagnosis, to name a few [3].

Most current QML research operates within the noisy intermediate-scale quantum (NISQ) regime, where devices are characterized by a limited number of qubits, restricted circuit depth, and susceptibility to noise [4]. As a result, many QML models have been developed and evaluated using classical simulators that emulate the quantum circuits. Frameworks such as PennyLane [5], Qiskit [6], TorchQuantum [7], and TensorFlow Quantum [8] were introduced to enable the integration of classical preprocessing with parameterized quantum circuits (PQCs). This integration facilitates end-to-end training through differentiable programming. Within these frameworks, variational quantum circuits (VQCs) employing parameterized rotational gates, data embedding strategies, and entanglement mechanisms have become standard due to their expressivity and compatibility with current quantum hardware constraints.

Recent research in QML has explored PQCs for classification and regression tasks, where classical data are encoded into quantum states and processed through trainable quantum operations [9], [10], [11]. These circuits can take different architectural forms, commonly referred to as *ansätze*. A widely used example is the hardware-efficient *ansatz*, which employs alternating layers of parameterized single-qubit rotations and entangling gates compatible with near-term quantum devices [12], [13]. Other forms include

problem-inspired *ansätze*, designed around the structure of a specific learning or physical problem [14], and data re-uploading circuits, which repeatedly encode the classical inputs across multiple layers to increase expressivity [15].

Under NISQ constraints, the choice of encoding strategy, circuit structure, and feature mapping plays a central role in model performance, and shallow entangling circuits are often preferred because they reduce the circuit depth and support stable training. Experimental and simulation-based studies have further shown that compact quantum models combined with classical preprocessing can provide a competitive performance on small- to medium-scale learning tasks. In particular, compressing high-dimensional classical features before quantum encoding has proven effective for reducing parameter count while preserving predictive performance [10]. Therefore, in this paper, we adopt a shallow hardware-efficient design based on R_X , R_Y , and R_Z rotations, followed by a linear chain of CNOT gates. This design is trainable, compatible with PennyLane, and easy to analyze in a tutorial-style text classification setting.

A particularly important application domain of QML is NLP [16], where tasks such as sentiment analysis [17], document classification [18], and subjectivity detection [19] involve high-dimensional and semantically-rich representations. Recent studies have indicated that QNNs provide promising structural advantages for such tasks, where embedding strategies such as amplitude encoding, which maps the normalized feature vectors directly to quantum state amplitudes, and angle encoding, which represents the features as rotation angles applied to quantum gates, enable efficient and expressive representations of linguistic patterns [20], [21].

Variational quantum circuits have shown promising performance in capturing nonlinear feature relationships within compact learning architectures [22]. For example, Caro et al. [23] and Ruskanda et al. [24] reported that compact quantum models can achieve competitive performance relative to classical networks in low-data regimes. Additionally, Yu et al. [25] demonstrated that quantum recurrent NNs formulated using VQCs can model linguistic dependencies effectively for low-resource text classification. These studies suggest that quantum circuits provide a flexible framework for exploring compact learning models under constrained computational environments.

Motivated by these advances, this paper presents a fully-reproducible quantum neural architecture for text classification that is minimal, interpretable, and feasible within the NISQ limitations. Our model uses term frequency-inverse document frequency (TF-IDF) for feature extraction, followed by a lightweight classical projection layer to compress inputs. The compressed features are then processed by a VQC consisting of four qubits and two layers, to perform quantum feature transformation by mapping classical features into quantum states through parameterized gate operations. Subsequent variational rotations and entanglement induce nonlinear transformations, enabling the extraction of expressive feature representations via quantum measurements.

In particular, the proposed framework uses a quantum-classical learning pipeline in which high-dimensional textual representations are first compressed through a lightweight classical projection layer and subsequently encoded into a parameterized VQC using angle embedding. This compact design provides a controlled setting for studying the role of quantum data encoding and circuit structure in NLP tasks. As the main architecture is trained end-to-end, the contribution of the downstream quantum head is further examined through an ablation study in which the classical compressor is frozen and multiple classifier heads are evaluated on the same shared low-dimensional representation.

We evaluate the proposed architecture across five benchmark datasets: movie reviews (MR) [26], subjectivity (SUBJ) [26], Stanford Sentiment Treebank (SST) [27], customer reviews (CR) [28] and the multi-perspective question answering (MPQA) [29] to analyze its performance, generalization behavior, and architectural efficiency. Our experimental results, supported by extensive ablation studies, show that even with a small quantum configuration, the proposed model achieves a competitive performance across all evaluated datasets and delivers the strongest overall results on CR, SUBJ and MPQA datasets, suggesting that the compact quantum-enhanced classifier heads can be effective in low-dimensional text classification pipelines.

Additional comparative experiments with compact classical models operating on the same compressed representation further allow a direct comparison of classifier heads within a shared low-dimensional feature space. To the best of our knowledge, in contrast to many existing studies where VQCs are combined with heavy deep learning components such as transformer encoders [18], [30], recurrent quantum models [25], convolutional layers [31], or quantum attention mechanisms [32], this work intentionally focuses on architectural simplicity. Instead of increasing depth or relying on large-scale pre-training, we study the role of the quantum layer by coupling a TF-IDF representation with a shallow variational circuit using a small number of qubits. This design enables a clearer understanding of how compact quantum circuits behave in NLP text classification and provides a controlled setting for analyzing the role of the VQC within the overall learning pipeline. The experimental results demonstrate that shallow VQCs combined with effective classical feature compression can yield a competitive performance on several benchmark datasets, supporting a minimal and reproducible framework for studying QNNs in NLP.

The key contributions of this paper are as follows:

- Provide a clear and structured presentation of QC concepts relevant to QML, including qubits, quantum gates, entanglement, measurement, and quantum data encoding.
- Introduce a compact quantum neural architecture compatible with PennyLane, featuring efficient TF-IDF compression and angle-encoded quantum embeddings.
- Conduct comprehensive experiments across five NLP benchmarks with detailed evaluations of accuracy,

F1-score, and area under the receiver operating characteristic curve (AUC-ROC).

- Provide a systematic comparison between the proposed QNN architecture and classical baselines, including compact multilayer perceptron (MLP) models operating on the same compressed feature representation.
- Conduct extensive ablation studies examining the effects of feature dimension, number of qubits, VQC depth, and entanglement type, allowing us to analyze how circuit configuration influences classification performance.

Table 1 provides a summary of the abbreviations and the remainder of this paper is organized as follows. Section II presents the conceptual background of QC and QML relevant to this work. Section III describes our proposed QNN text classification architecture. Section IV details the experimental setup, datasets, and evaluation metrics. Section V reports and discusses the results, including ablation studies and additional analyses. Finally, Section VI concludes the paper and outlines the future research directions.

TABLE 1. Abbreviations used in this paper.

CC	Classical Computing
QC	Quantum Computing
QNN	Quantum Neural Network
NN	Neural Network
NISQ	Noisy Intermediate-scale Quantum
NLP	Natural Language Processing
PQC	Parameterized Quantum Circuit
VQC	Variational Quantum Circuit
QML	Quantum Machine Learning
TF-IDF	Term Frequency Inverse Document Frequency
MLP	Multilayer Perceptron
Qubit	Quantum Bit
CNOT	Controlled-NOT
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
MR	Movie Review
CR	Customer Review
SUBJ	Subjectivity
SST	Stanford Sentiment Treebank
MPQA	Multi-perspective Question Answering

II. CONCEPTUAL BACKGROUND

This section reviews the fundamental concepts, including qubits, quantum gates, entanglement, and measurement, which originate from QC and are directly used in QML. As illustrated in Fig. 1, QC provides the primary concepts, while QML builds upon them through a structured learning pipeline. In general, QML models follow a three-stage process consisting of data encoding, quantum computation, and measurement [33]. In the encoding stage, classical data are mapped into quantum states using feature embedding strategies. The quantum computation stage applies VQCs to transform these states through unitary operations and entanglement, while the measurement stage extracts classical information from the quantum system, typically through expectation values of observables.

Based on this framework, several QML approaches have been developed, including QNNs, quantum kernel methods, quantum clustering, and quantum generative models, as shown in Fig. 1. However, in this work, we focus on QNNs,

where VQCs are integrated with the classical learning components. Accordingly, the following sections also emphasize VQCs, data encoding strategies, and measurement operations that underpin the proposed model.

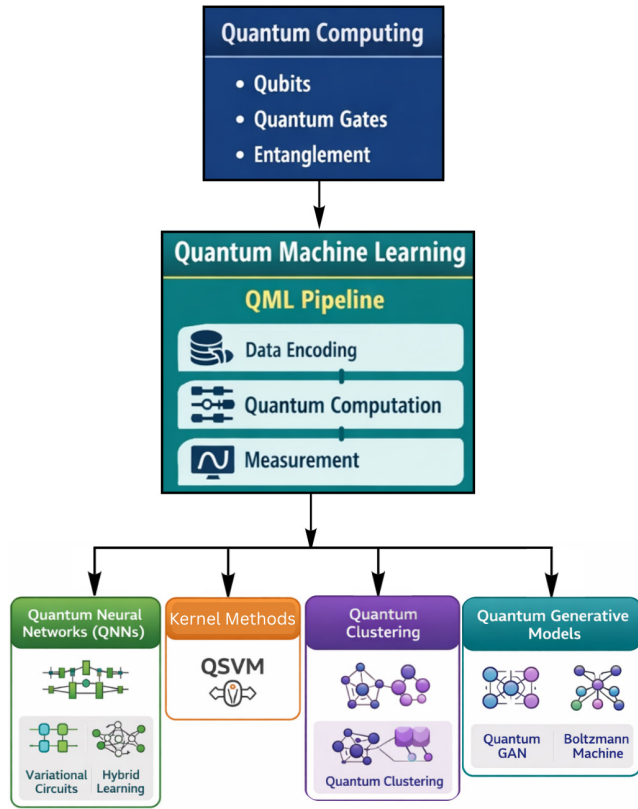


FIGURE 1. Overview of the relationship between QC, QML, and common QML approaches. Generally, QC provides the foundational concepts such as qubits, gates, and entanglement and QML builds upon these concepts through a three-stage pipeline consisting of data encoding, quantum computation, and measurement [33].

A. QUANTUM BITS

In CC, the information is represented using *bits*, which can exist in one of two possible states: 0 or 1. These states typically correspond to physical conditions such as low or high voltage, false or true, off or on. In QC, however, the fundamental unit of information is the qubit [34]. Unlike the classical bit, a qubit behaves according to the principles of quantum mechanics, particularly the concepts of superposition and probability amplitudes [35].

Mathematically, the state of a qubit can be described as a linear combination or superposition of two basis states denoted as $|0\rangle$ and $|1\rangle$. It is expressed as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where $|\psi\rangle$ (read as “ket psi”) represents the state of the qubit, and α and β are complex numbers known as probability amplitudes. It is important to note that α and β themselves are not probabilities; rather, their squared magnitudes, $|\alpha|^2$ and $|\beta|^2$, correspond to the actual probabilities of measuring the qubit in states $|0\rangle$ and $|1\rangle$, respectively. Because the total

probability must be equal to one, the amplitudes satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. This ensures that when a measurement is performed, the qubit collapses into a definite state either $|0\rangle$ or $|1\rangle$ with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively.

1) UNDERSTANDING THE SUPERPOSITION

The concept of superposition makes QC fundamentally different and more powerful than CC. A qubit in superposition means it is not just in state 0 or 1, but rather in both states simultaneously, with certain probabilities associated with each [36]. For example, if $\alpha = \frac{1}{\sqrt{2}}$ and $\beta = \frac{1}{\sqrt{2}}$, the qubit is represented as $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$, meaning that there is an equal probability (50%) of measuring it as 0 or 1. This unique ability to exist in multiple states at once enables *quantum parallelism*, allowing quantum computers to evaluate many possibilities simultaneously and perform certain computations far more efficiently than classical computers do.

2) QUANTUM STATE VISUALIZATION VIA BLOCH SPHERE

A helpful way to visualize the state of a qubit is through the *Bloch sphere*, a three-dimensional geometric representation that captures all physically distinct pure qubit states. The Bloch sphere is a unit sphere, where each point on its surface corresponds uniquely to a qubit state, up to an unobservable global phase. As illustrated in Fig. 2, the vertical z -axis is aligned with the computational basis. The north pole corresponds to the basis state $|0\rangle$, while the south pole corresponds to $|1\rangle$. Any other point on the surface of the sphere represents a valid superposition of these two states.

The state of the qubit is depicted by a vector originating from the center of the sphere and terminating on its surface (shown as the red arrow in Fig. 2). This state vector is fully specified (for pure states) by two angles in spherical coordinates. The polar angle $\theta \in [0, \pi]$ measures the inclination of the state vector from the positive z -axis and determines the relative weights or amplitudes of the basis states $|0\rangle$ and $|1\rangle$. The azimuthal angle $\phi \in [0, 2\pi)$ specifies a rotation about the z -axis and encodes the relative phase between the basis amplitudes.

Using this parametrization, an arbitrary single-qubit pure state can be written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle,$$

where the coefficients $\alpha = \cos(\theta/2)$ and $\beta = e^{i\phi}\sin(\theta/2)$ satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. The magnitudes $|\alpha|^2$ and $|\beta|^2$ correspond to the probabilities of measuring the qubit in the $|0\rangle$ and $|1\rangle$ states, respectively, while the relative phase ϕ influences interference effects and the qubit’s evolution under quantum operations.

The three orthogonal axes of the Bloch sphere are commonly referred to as the Pauli axes. The z -axis corresponds to measurement in the computational basis, while the x - and y -axes correspond to measurements in the $|+\rangle$, $|-\rangle$

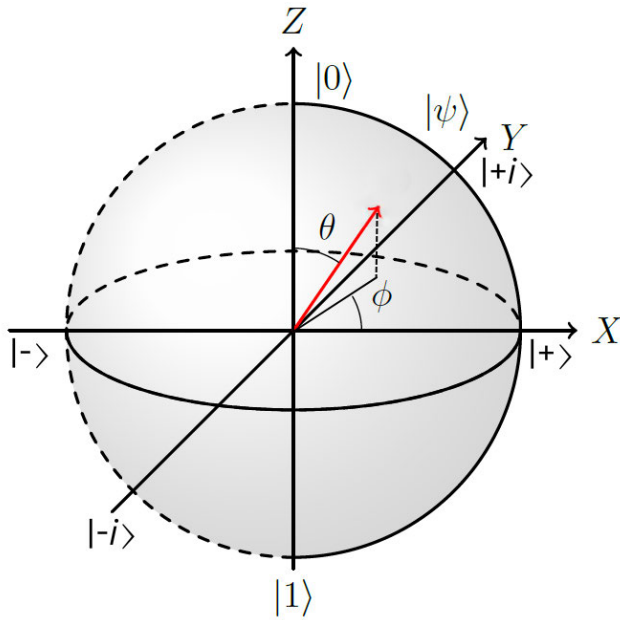


FIGURE 2. Bloch sphere representation of a single-qubit pure state. The north and south poles correspond to the computational basis states $|0\rangle$ and $|1\rangle$, respectively. The x -, y -, and z -axes correspond to the Pauli X , Y , and Z axes. A qubit state is represented by a unit vector originating at the center of the sphere and terminating on its surface, parameterized by the polar angle θ (from the $+z$ -axis) and the azimuthal angle ϕ (in the x - y plane). Arrowheads indicate axes pointing away from the viewer, while plain lines lie in the plane of the page.

and $|+i\rangle$, $|-i\rangle$ bases, respectively. The points on the equator ($\theta = \pi/2$) therefore represent equal-probability superposition states that differ only in their relative phase.

Single-qubit quantum gates act geometrically as rotations of the state vector about one of the Bloch-sphere axes. Since all physically valid single-qubit gates are represented by unitary operators, these rotations preserve the length of the state vector, which ensures the quantum state remains on the surface of the sphere throughout its evolution. This geometric viewpoint provides an intuitive understanding of quantum dynamics, where the computation corresponds to controlled rotations on the Bloch sphere rather than transitions between the discrete classical states. While a classical bit occupies one of two discrete states at a time, a qubit spans a continuous set of superposition states characterized by both amplitude and relative phase. When extended to multi-qubit systems, this geometric freedom leads to an exponential growth of the accessible state space, with an n -qubit system spanning 2^n basis states simultaneously, forming the basis of quantum parallelism and potential quantum speedups in tasks such as integer factorization, database search, and quantum system simulation [37].

B. QUANTUM GATES AND ENTANGLEMENT

Quantum gates are the fundamental operations that manipulate the state of qubits, similar to how classical logic gates operate on bits. Unlike classical bits, qubits are represented as two-dimensional complex vectors. Hence, by using the

computational basis states $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, a general qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ can be written as

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

Single-qubit quantum gates are linear operators that act on these vectors and can often be interpreted as rotations on the Bloch sphere. One of the most fundamental gates is the Pauli- X gate, sometimes referred to as the quantum NOT gate, which flips the state of a qubit. Geometrically, the Pauli- X gate corresponds to a π rotation about the x -axis of the Bloch sphere. Its matrix representation is given by

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

and applying it to the qubit state $|\psi\rangle$ yields

$$X|\psi\rangle = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}.$$

This operation exchanges the amplitudes of $|0\rangle$ and $|1\rangle$, meaning that $|0\rangle$ becomes $|1\rangle$ and $|1\rangle$ becomes $|0\rangle$.

The Pauli- Y gate can be interpreted geometrically as a π rotation about the y -axis on the Bloch sphere and introduces a phase factor of i or $-i$ depending on the state. Its matrix form is given by

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix},$$

and by acting on the vector representation of a qubit state $|\psi\rangle$, it produces

$$Y|\psi\rangle = \begin{bmatrix} -i\beta \\ i\alpha \end{bmatrix}.$$

Thus, the Pauli- Y gate not only flips the amplitudes but also introduces a relative phase shift, which is essential for interference effects in quantum algorithms.

The Pauli- Z gate applies a phase flip to the $|1\rangle$ component while leaving $|0\rangle$ unchanged. Similarly, from a geometric perspective, the Pauli- Z gate corresponds to a π rotation about the z -axis of the Bloch sphere. Its matrix representation is expressed as

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

and when applied to a general qubit state $|\psi\rangle$, it yields

$$Z|\psi\rangle = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}.$$

Here, the $|0\rangle$ amplitude remains unchanged, while the $|1\rangle$ amplitude acquires a negative sign, corresponding to a phase flip of π .

Another frequently used single-qubit gate is the Hadamard or H gate, which creates an equal superposition of $|0\rangle$ and $|1\rangle$, enabling quantum parallelism. Its matrix form is defined by

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

and when applied to a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, it produces $H|\psi\rangle = \frac{\alpha+\beta}{\sqrt{2}}|0\rangle + \frac{\alpha-\beta}{\sqrt{2}}|1\rangle$. This transformation allows a qubit that is initially in a basis state to enter a superposition state, forming the basis for quantum interference and parallelism.

In addition to single-qubit gates, multi-qubit gates operate on two or more qubits, and are essential for creating quantum entanglement, which is a phenomenon with no classical counterpart [38]. One of the most widely used multi-qubit gates is the controlled-NOT (CNOT) gate [39], which acts on a pair of qubits by designating one as the control and the other as the target. If the control qubit is in the state $|1\rangle$, the CNOT gate flips the state of the target qubit; whereas if the control qubit is in $|0\rangle$ state, the target qubit remains unchanged [40]. In matrix form, when acting on the standard two-qubit basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the CNOT gate is represented as follows:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Applied to a general two-qubit state $|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$, the CNOT gate transforms it into

$$\text{CNOT}|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|11\rangle + \delta|10\rangle$$

This operation is fundamental for generating entanglement, as it can create correlations between the qubits that have no classical analogue [41]. In practical quantum circuits, the CNOT gates are often applied after the layers of single-qubit rotations to enhance the expressivity of the quantum system [30] and enable it to capture complex feature interactions in quantum algorithms.

In contrast to the CNOT gate, the IsingZZ ($R_{ZZ}(\theta)$) gate is another standard two-qubit entangling operation which is widely used in variational quantum algorithms. It is derived from the Ising Hamiltonian, a model that describes pairwise interactions between spins or qubits. In its simplest two-qubit form, the Ising Hamiltonian is expressed as

$$H_{\text{Ising}} = Z \otimes Z,$$

which characterizes how the energy of the system depends on whether the qubits are aligned or anti-aligned in the computational Z-basis. Here, Z is the Pauli-Z operator acting on each qubit. Exponentiating this Hamiltonian gives the IsingZZ unitary:

$$\text{IsingZZ}(\theta) = e^{-i(\theta/2)Z \otimes Z},$$

In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the matrix form of this gate is

$$\text{IsingZZ}(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 & 0 & 0 \\ 0 & e^{i\theta/2} & 0 & 0 \\ 0 & 0 & e^{i\theta/2} & 0 \\ 0 & 0 & 0 & e^{-i\theta/2} \end{bmatrix}.$$

Here, the states $|00\rangle$ and $|11\rangle$ are referred to as *aligned*, since both qubits share the same eigenvalue of the Z operator,

while $|01\rangle$ and $|10\rangle$ are *anti-aligned*, having opposite Z eigenvalues. The IsingZZ gate applies one phase factor $e^{-i\theta/2}$ to aligned states, and the opposite phase factor $e^{i\theta/2}$ to anti-aligned states. This conditional phase evolution becomes nontrivial when the qubits are in superposition, enabling the generation of entanglement. Because the entangling strength is directly controlled by the parameter θ , the IsingZZ gate offers a tunable alternative to the CNOT gate in VQCs and Ising-type Hamiltonian simulations [42].

C. QUANTUM DATA ENCODING

Unlike classical NNs, quantum circuits operate on quantum states rather than numerical feature vectors. Therefore, classical data must be explicitly encoded into quantum states before it can be processed by a quantum model. This translation step is fundamental in QC, as quantum operations act on the amplitudes and relative phases of quantum states rather than directly on classical values.

In QML, this mapping from classical data to quantum states is referred to as quantum data encoding. This encoding step is a core component of the QNN architectures, where classical features must be transformed into quantum states before being processed by the VQCs. The choice of encoding strategy plays a central role in determining how effectively a quantum model can represent, learn from, and generalize the classical inputs [21]. In recent literature, a variety of encoding schemes have been proposed, including basis encoding, superposition encoding, amplitude encoding, and angle encoding [20], [43]. Each approach involves distinct trade-offs in terms of representational capacity, circuit depth, and implementation complexity.

Among these methods, *angle encoding* (or *angle embedding*) is one of the most widely used techniques in VQCs. It maps classical features represented by real values to quantum states by encoding each feature as a rotation angle implemented through a single-qubit rotation gate on the Bloch sphere [21]. This takes advantage of quantum gates that perform rotations around different axes of the Bloch sphere, such as $R_X(\theta)$, $R_Y(\theta)$, and $R_Z(\theta)$.

For example, a rotation about the Y -axis is given by

$$R_Y(\theta) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{bmatrix},$$

which rotates the qubit state vector by an angle θ about the y -axis. Applying this gate to the initial state $|0\rangle$ yields

$$R_Y(\theta)|0\rangle = \cos(\frac{\theta}{2})|0\rangle + \sin(\frac{\theta}{2})|1\rangle.$$

Because the $R_Y(\theta)$ operator has a purely real matrix representation in the computational basis, its action introduces no relative complex phase between $|0\rangle$ and $|1\rangle$, and the resulting state corresponds to an azimuthal angle $\phi = 0$ on the Bloch sphere. Here, the classical feature value is encoded directly as the rotation angle θ , which determines the probability amplitudes of measuring $|0\rangle$ or $|1\rangle$. While the input feature

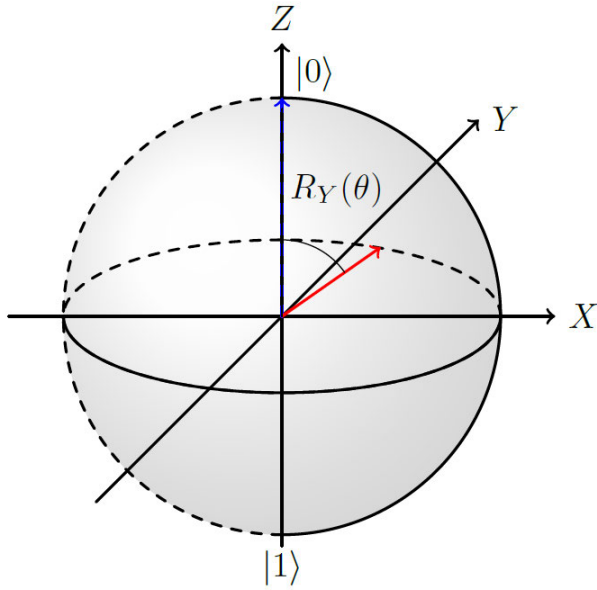


FIGURE 3. Visualization of angle embedding on the Bloch sphere. The qubit, initially in the $|0\rangle$ state at the north pole (shown as the blue arrow), is rotated about the y -axis by an angle θ , producing the superposition $R_Y(\theta)|0\rangle = \cos(\frac{\theta}{2})|0\rangle + \sin(\frac{\theta}{2})|1\rangle$. The rotation angle θ corresponds to a classical feature value. Because the rotation is performed about the y -axis, the azimuthal angle remains $\phi = 0$, and the state vector evolves along the x - z plane.

itself is not normalized, the resulting quantum state remains normalized due to the unitary nature of the rotation operation.

In a multi-qubit setting, each feature is encoded on a separate qubit, establishing a direct correspondence between the reduced input dimension and the number of qubits. As illustrated in Fig. 3, angle embedding can be visualized on the Bloch sphere as a rotation of the qubit state vector from the north pole (corresponding to $|0\rangle$) toward the south pole (corresponding to $|1\rangle$), with the trajectory confined to the x - z plane. This rotation transforms the qubit from a computational basis state into a superposition state within its two-dimensional Hilbert space, which is the complex vector space spanned by $\{|0\rangle, |1\rangle\}$.

In the proposed QNN model, angle embedding is implemented using PennyLane's `AngleEmbedding` template, which applies R_Y rotations to all qubits in parallel according to the input feature vector. Here, n_q denotes the number of qubits in the quantum circuit and corresponds to the dimensionality of the encoded feature vector. This operation is defined as

```
pennylane.templates.  
AngleEmbedding(inputs, wires=range(n_q),  
rotation='Y').
```

Each element of the input vector, obtained after classical dimensionality reduction through a linear projection of TF-IDF features, specifies the rotation angle applied to the corresponding qubit. As a result, the VQC used in this paper simultaneously encodes the compressed classical

information across all qubits, enabling expressive quantum feature transformations with minimal circuit depth.

D. QUANTUM MEASUREMENT

In QNNs, measurement is the process of extracting the classical information from the quantum state after all parameterized gate operations have been applied. This measurement stage forms the final step of the QNN inference, where the quantum states are mapped back to classical features for downstream prediction tasks. In this paper, the measurement is performed using the expectation value of the Pauli-Z operator on each qubit, defined as

$$\langle Z_i \rangle = \langle \psi | Z_i | \psi \rangle,$$

where Z_i denotes the Pauli-Z operator acting on the i^{th} qubit [44] and $|\psi\rangle$ represents the final quantum state produced by the variational circuit. In Dirac notation, the bra $\langle \psi |$ is the Hermitian conjugate (complex conjugate transpose) of the ket $|\psi\rangle$. This formulation ensures that the expectation value $\langle \psi | Z_i | \psi \rangle$ is a real-valued scalar, making it suitable for gradient-based optimization in quantum-classical learning frameworks [45].

The expectation value $\langle Z_i \rangle$ lies in the interval $[-1, 1]$ and reflects the alignment of the qubit state along the Z -axis of the Bloch sphere. A value of $+1$ corresponds to the qubit being in the computational basis state $|0\rangle$, while a value of -1 corresponds to the state $|1\rangle$. Intermediate values indicate a probabilistic mixture of these basis states. Equivalently, the Pauli-Z expectation value can be expressed as

$$\langle Z_i \rangle = P_i(0) - P_i(1),$$

where $P_i(0)$ and $P_i(1)$ denote the probabilities of measuring the i^{th} qubit in the states $|0\rangle$ and $|1\rangle$, respectively.

The Pauli-Z operator is commonly used in VQCs due to its direct compatibility with computational-basis measurements, numerical stability, and ease of integration with classical NN layers [46]. While alternative observables may also be considered, the Pauli-Z measurements provide a simple and expressive readout mechanism, as the variational circuit can learn to rotate the task-relevant information into the measurement basis through parameterized rotations and entanglement [47]. In our proposed QNN, parameterized rotation gates such as $R_X(\theta)$, $R_Y(\theta)$, and $R_Z(\theta)$ are used for data encoding and trainable variational transformation purpose, whereas the Pauli-Z operator is used for measurement.

III. ARCHITECTURE OF THE QNN TEXT CLASSIFICATION MODEL

The overall flow diagram of our QNN pipeline is illustrated in Fig. 4. Here, the classical front-end performs the TF-IDF vectorization and dimensionality reduction, while the quantum backend executes the VQC that encodes, entangles, and measures the qubit states to extract the quantum-enhanced features for the final classification. The detailed VQC structure is depicted in Fig. 5. The VQC used for classification consists of an angle-encoding layer that maps

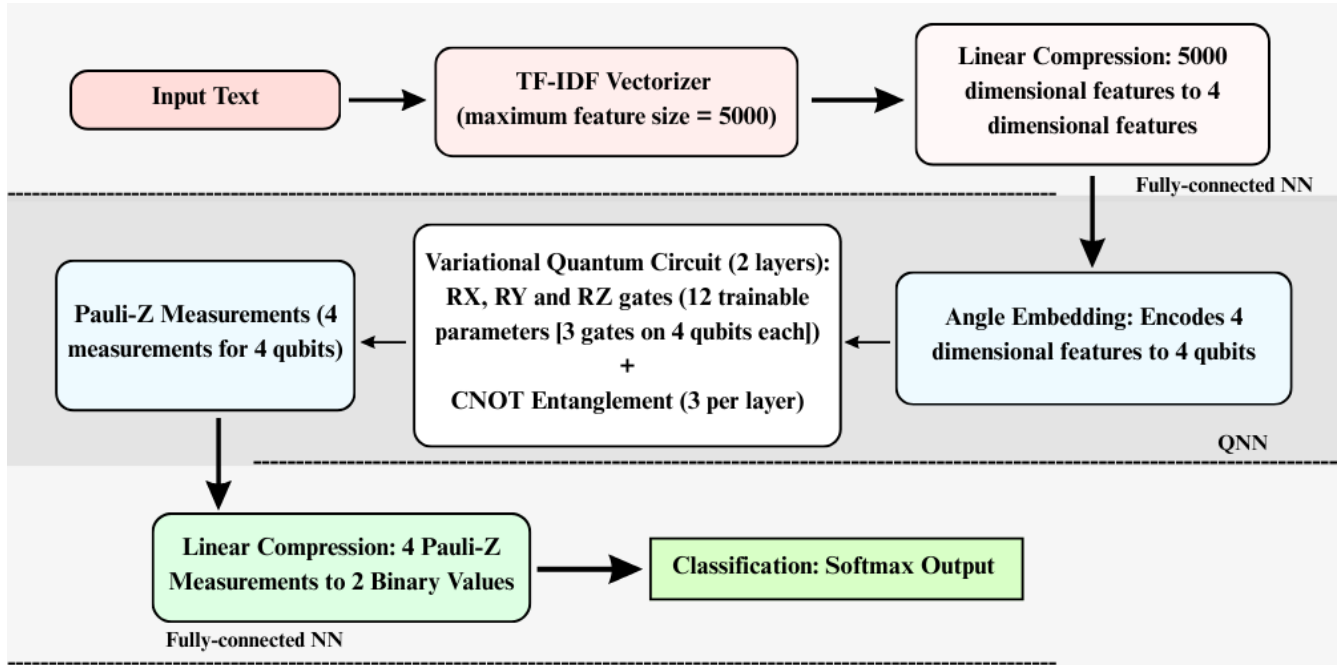


FIGURE 4. Flow diagram of our proposed QNN model. The core component of the pipeline is the VQC, which is further illustrated in Fig. 5.

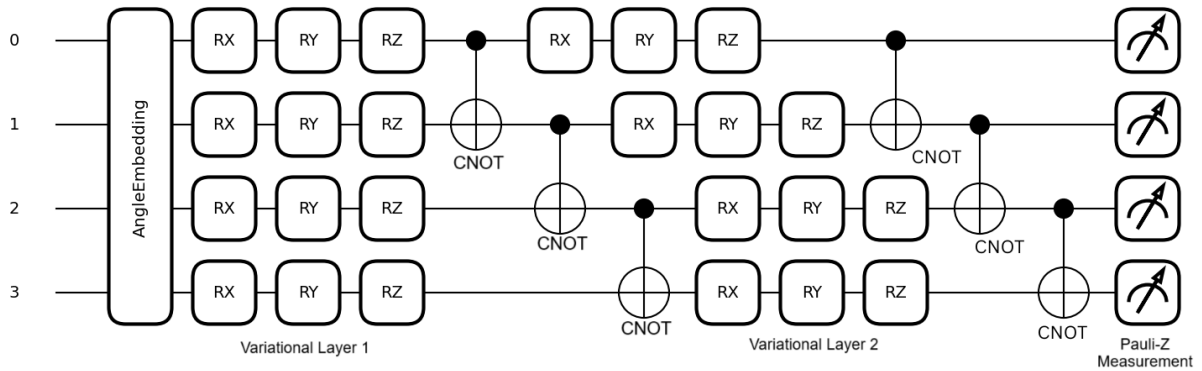


FIGURE 5. An illustration of the VQC consisting of four qubits initialized in the $|0\rangle$ state, followed by an Angle Embedding layer that encodes classical data as rotation angles. Layers of parameterized single-qubit rotations (R_X , R_Y , and R_Z) and CNOT gates create entanglement and enable complex feature transformations. The circuit ends with the Pauli-Z measurements on all qubits, producing expectation values that serve as quantum features for the classical network. This configuration is chosen based on detailed ablation studies in Section V-A4.

the compressed TF-IDF features to qubit rotation angles, followed by two layers of parameterized single-qubit rotations and entangling gates. The circuit parameters are optimized during training using classical gradient-based optimization, and the expectation values of Pauli-Z measurements serve as the quantum features for the final classifier.

A. CLASSICAL FEATURE EXTRACTION AND DATA COMPRESSION

To illustrate how our proposed model processes the text data, consider the sentence “that’s far too tragic to merit such superficial treatment,” taken from the CR dataset [28] used for sentiment classification. First, the sentence is converted into a numerical vector using the TF-IDF method [48], which

measures the relative importance of each word in the text based on its frequency across the corpus. The maximum feature size is set to 5000, determined experimentally in Section V-A2, after observing an improved performance across datasets.

This 5000-dimensional TF-IDF vector $\mathbf{x} \in \mathbb{R}^{5000}$ is then passed through a fully-connected linear layer that compresses the high-dimensional input into a four-dimensional representation aligned with the number of qubits in the quantum circuit. The transformation is defined as:

$$\mathbf{h}_1 = \tanh(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1),$$

where $\mathbf{W}_1 \in \mathbb{R}^{n_q \times 5000}$, $\mathbf{b}_1$ is the bias vector, and $n_q = 4$ denotes the number of qubits. The tanh activation ensures

that the resulting feature values are normalized within the range $[-1, 1]$, suitable for encoding as quantum rotation angles during state preparation.

B. DATA ENCODING

Each of these four compressed features is then mapped to a qubit through the *AngleEmbedding* operation as shown in Fig. 3. As discussed earlier, each feature value is interpreted as a rotation angle on the Bloch sphere [21], typically applied through an $R_Y(\theta_i)$ gate, where θ_i is the feature corresponding to qubit i . As a result, the classical information is embedded in quantum amplitudes, which allows the circuit to represent a superposition of all possible states weighted by these rotations. This step bridges the classical and quantum domains, converting the numeric sentiment features into the quantum-encoded information.

C. VARIATIONAL QUANTUM CIRCUIT AND MEASUREMENT

As shown in Fig. 5, the angle embedded qubits then pass through two layers of VQCs. In each variational layer, every qubit is processed by a sequence of three parameterized single-qubit rotations (R_X , R_Y , and R_Z). The parameters $\theta_{i,1}$, $\theta_{i,2}$, and $\theta_{i,3}$ are trainable real-valued angles associated with the i^{th} qubit, corresponding to rotations around the X , Y , and Z axes, respectively. These rotations define the local unitary which is expressed by

$$U_i(\theta_i) = R_Z(\theta_{i,3})R_Y(\theta_{i,2})R_X(\theta_{i,1}),$$

where $\theta_i \in \mathbb{R}^3$. For a circuit with n_q qubits, the full variational operation is given by

$$U(\theta) = \prod_{i=1}^{n_q} U_i(\theta_i),$$

which applies independent rotations to all qubits in parallel.

To introduce the inter-qubit correlations, a linear chain of CNOT gates is applied after the rotation layer:

$$\text{CNOT}(q_i, q_{i+1}), \quad i = 1, \dots, n_q - 1,$$

where q_i and q_{i+1} denote the control and target qubits, respectively. While the rotation layer performs the local feature transformations, the CNOT operations entangle the neighboring qubits, enabling the circuit to capture the higher-order feature interactions with a compact parameterization [22].

These operations collectively form the variational portion of the model, introducing both trainable parameters and qubit entanglement to capture the complex, nonlinear relationships between features. After all variational layers, the circuit measures the expectation values of the Pauli-Z operators to extract the quantum features. For the i^{th} qubit, this expectation value is given by

$$z_i = \langle 0|U^\dagger Z_i U|0\rangle, \quad i = 1, \dots, n_q,$$

where Z_i denotes the Pauli-Z operator acting on the i^{th} qubit, and z_i is the corresponding real-valued measurement outcome. Here, U represents the overall unitary transformation implemented by the parameterized VQCs, comprising the data encoding, variational rotation, and entanglement layers, while $U^\dagger$ denotes its Hermitian adjoint, i.e., the conjugate transpose. The resulting expectation values z_1, z_2, z_3, z_4 are real numbers in the range $[-1, 1]$, forming a four-dimensional quantum output vector $\mathbf{z} \in \mathbb{R}^4$.

D. CLASSICAL OUTPUT LAYER AND PREDICTION

These quantum outputs are then passed back to the classical domain through a final linear layer for binary sentiment classification:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_2 \mathbf{z} + \mathbf{b}_2),$$

where $\mathbf{W}_2 \in \mathbb{R}^{2 \times n_q}$ and $\mathbf{b}_2 \in \mathbb{R}^2$ denote the trainable weight matrix and bias vector of the output classifier, respectively. Unlike $\mathbf{W}_1$, which projects high-dimensional TF-IDF features into a low-dimensional space suitable for quantum encoding, $\mathbf{W}_2$ operates on the quantum measurement outputs $\mathbf{z}$ and performs the final decision mapping from quantum features to class probabilities. Thus, $\mathbf{W}_1$ serves as a feature compression layer, while $\mathbf{W}_2$ functions as a task-specific classification layer. The softmax activation produces a probability distribution $\hat{\mathbf{y}} \in \mathbb{R}^2$ over two sentiment classes. This binary setting is adopted because the datasets used in this study have two labels negative sentiment (class 0) and positive sentiment (class 1).

If the resulting output vector from the classifier indicates a higher probability for class 0, the model interprets the review as negative, reflecting a dissatisfaction or criticism. In this case, the sentence “that’s far too tragic to merit such superficial treatment” would likely be assigned to class 0 because its linguistic cues “too tragic,” “superficial,” and the negative tone strongly suggest unfavorable sentiment. Conversely, a sentence expressing praise or satisfaction, such as “a wonderfully heartfelt performance,” would produce a higher probability for class 1, indicating a positive sentiment. This example demonstrates how our QNN model integrates quantum feature transformations with a classical decision-making to perform the nuanced sentiment classification tasks.

IV. EXPERIMENTAL SETUPS

In this section, we discuss the considered datasets and how they are processed for the proposed text classification task using our QNN model.

A. DATASETS

Our experiments are conducted using five widely recognized benchmark datasets for text classification, namely, MR, SST, SUBJ, CR, and MPQA. Each dataset is partitioned into training, validation, and test subsets following an 8:1:1 ratio, ensuring a consistent evaluation protocol across tasks. However, the SST dataset [27] follows a different configuration. We use the SST version provided through the Hugging Face repository [49], where the official test

labels are not released. To obtain a labeled test set for our analysis, we combine the available labeled training data and extract 1,821 examples as the test portion, resulting in a final split of 65,528 training samples, 872 validation samples, and 1,821 test samples. Each SST instance corresponds to a single sentence annotated as positive or negative, making the dataset well suited for evaluating the models on fine-grained sentiment classification using carefully curated annotations.

The MR dataset [26] contains 10,662 single-sentence movie reviews with balanced positive and negative labels. The SUBJ dataset [26] comprises 10,000 sentences labeled as subjective or objective, providing a benchmark for distinguishing personal opinions from factual statements. The CR dataset [28] consists of 3,772 short product reviews annotated for sentiment polarity, while the MPQA corpus [29] includes 10,603 sentences drawn from news articles, capturing the fine-grained subjective nuances and opinion expressions. The detailed statistics of these datasets are summarized in Table 2.

TABLE 2. Dataset statistics after training, validation, and test splits.

Dataset	Train	Validation	Test	Total	Labels
MR	8530	1065	1067	10662	pos/neg
SST	65528	872	1821	68221	pos/neg
SUBJ	8000	1000	1000	10000	sub/obj
CR	3024	364	384	3772	pos/neg
MPQA	8496	1035	1072	10603	pos/neg

B. TRAINING SETTINGS

All datasets are processed using our proposed QNN model, implemented in PyTorch and PennyLane. This model consists of a classical fully-connected layer that compresses the TF-IDF features to match the number of qubits, a VQC with 4 qubits and 2 layers employing Angle Embedding, trainable rotations, and entangling CNOT gates, followed by a final classical linear layer projecting quantum outputs to the two-class label space. For training, the Adam optimizer with a learning rate of 0.001 and cross-entropy loss are used, with a batch size of 32 over 10 epochs.

The training data are shuffled at each epoch, and the gradients are backpropagated through both the classical and quantum layers. The validation accuracy and F1-score are monitored at the end of every epoch. The test set is evaluated using the model parameters saved at the epoch that achieved the best validation performance, ensuring an optimal generalization. The test predictions are obtained by using the argmax of the final layer outputs, and the probabilities for area under the receiver operating characteristic curve (AUC-ROC) computation are derived from the softmax logits. The same training setup and hyperparameters are applied consistently across all the datasets to ensure a fair comparison.

C. EVALUATION METRICS

To quantitatively evaluate the performance of our proposed quantum model and its classical counterpart, we employ three key metrics, namely accuracy, F1-score, and AUC-ROC.

Accuracy measures the ratio of correctly predicted samples to the total number of samples and it is defined as

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}.$$

where TP , TN , FP , and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

F1-score represents the harmonic mean of precision and recall, providing a balanced assessment between FP and FN:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Finally, AUC-ROC quantifies the model's ability to distinguish between classes across all possible classification thresholds. It is computed as the area under the ROC curve, where the curve plots the true positive rate (TPR) against the false positive rate (FPR):

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}$$

It should be noted that TPR is mathematically equivalent to Recall, both defined as $\frac{TP}{TP + FN}$, and is referred to as TPR in the ROC analysis to emphasize threshold-based evaluation. A higher AUC-ROC value indicates a superior discriminative power. Together, these metrics offer a comprehensive evaluation of model accuracy, balance, and robustness.

V. RESULTS AND DISCUSSION

To evaluate the performance of our proposed QNN model, we compare it with several classical baselines (Table 3) while keeping the feature extraction pipeline and data splits identical. In addition to a larger classical NN, we include compact classical baselines that operate on the same four-dimensional compressed feature representation used as input to the QNN. Specifically, the classical NN (5000→128→2) represents a standard MLP with a 128-unit hidden layer, whereas the smaller NN (5000→4→2) operates directly on the four-dimensional compressed features aligned with the four-qubit input of the QNN. Logistic regression is also included as a lightweight linear baseline using the same compressed representation. Table 3 presents a detailed comparison of classification results across the five benchmark sentiment analysis datasets.

Overall, our proposed QNN demonstrates a competitive performance compared with that of both the larger and compact classical NN baselines. In particular, the QNN achieves the best results on the CR, SUBJ and MPQA datasets in terms of accuracy, F1-score and AUC-ROC. For example, on the CR dataset, the QNN achieves an accuracy of 0.8307, outperforming both classical NN baselines and logistic regression. Similarly, on SUBJ, the QNN attains the highest accuracy and F1-score of 0.9170 and 0.9130, respectively, while maintaining the best AUC-ROC score among all models. On MPQA, the QNN again provides the strongest overall performance across the evaluated metrics.

TABLE 3. Performance comparison of the proposed QNN and classical baselines on five text classification datasets. All models use TF-IDF features (maximum vocabulary size of 5000) and identical data splits. Classical NN (5000→128→2) denotes an MLP with a 128-unit hidden layer, while Classical NN (5000→4→2) and logistic regression use the same four-dimensional compressed feature representation, aligned with the four-qubit input of the QNN. Results are reported in terms of Accuracy, F1-score, and AUC-ROC. The best result for each dataset is shown in bold.

Dataset	Model	Accuracy	F1-Score	AUC-ROC
MR	QNN (Proposed)	0.7844	0.7843	0.8659
	Classical NN (5000→4→2)	0.7826	0.7825	0.8610
	Classical NN (5000→128→2)	0.7554	0.7554	0.8335
	Logistic Regression	0.7882	0.7881	0.8660
CR	QNN (Proposed)	0.8307	0.8292	0.8952
	Classical NN (5000→4→2)	0.8151	0.8119	0.8934
	Classical NN (5000→128→2)	0.7969	0.7952	0.8666
	Logistic Regression	0.8255	0.8242	0.8943
SUBJ	QNN (Proposed)	0.9170	0.9130	0.9690
	Classical NN (5000→4→2)	0.9050	0.9050	0.9673
	Classical NN (5000→128→2)	0.8930	0.8930	0.9572
	Logistic Regression	0.8980	0.8980	0.9574
MPQA	QNN (Proposed)	0.8731	0.8706	0.9271
	Classical NN (5000→4→2)	0.8675	0.8657	0.9142
	Classical NN (5000→128→2)	0.8619	0.8613	0.9095
	Logistic Regression	0.8657	0.8645	0.9177
SST	QNN (Proposed)	0.8764	0.8759	0.9462
	Classical NN (5000→4→2)	0.8523	0.8512	0.9339
	Classical NN (5000→128→2)	0.8951	0.8953	0.9601
	Logistic Regression	0.8561	0.8557	0.9338

TABLE 4. Comparison of trainable parameters for the classical and quantum models.

Layer / Component	QNN (proposed) (5000→4→Quantum→2)	Logistic Regression or Classical NN (5000→4→2)	Classical NN (5000→128→2)
Compression Layer	$5000 \times 4 + 4 = 20,004$	$5000 \times 4 + 4 = 20,004$	$5000 \times 128 + 128 = 640,128$
Quantum Layer	$2 \times 4 \times 3 = 24$	—	—
Output Layer	$4 \times 2 + 2 = 10$	$4 \times 2 + 2 = 10$	$128 \times 2 + 2 = 258$
Total Parameters	20,038	20,014	640,386

On the MR dataset, the results are closely matched across models. Logistic regression achieves the highest accuracy of 0.7882, while the QNN performs comparably with an accuracy of 0.7844 and similar AUC-ROC values. This suggests that the task can already be well modeled using relatively simple classifiers operating on TF-IDF representations.

The larger classical NN achieves the best overall performance on the SST dataset. This is reasonable given the large training size of the SST dataset, which may benefit from models with a higher number of parameters. The QNN nevertheless remains competitive with the compact baselines, while the smaller classical models underperform on this benchmark.

These results indicate that when the classifier capacity is matched through the shared four-dimensional compressed feature representation, the QNN behaves comparably to classical models operating on the same compact feature space. At the same time, the QNN achieves the best results on several datasets despite its shallow circuit structure and small number of trainable parameters as shown in Table 4. This suggests that the quantum circuit can serve as a competitive low-dimensional classifier head within the proposed pipeline.

To further understand the behavior of our proposed model under different architectural choices, we conduct a series of ablation studies in Section V-A, analyzing the impact of the

feature compression stage, TF-IDF feature dimension, qubit count, and circuit choice and VQC layer depth on the overall classification performance.¹

A. ABLATION STUDY

In the earlier part of this section, the best results achieved by our proposed QNN model across multiple benchmark datasets were presented. These results were obtained after conducting extensive experiments to explore the effects of key architectural and hyperparameter choices. In particular, the ablation study performed here is meant to examine how the classical feature compression stage, the TF-IDF feature dimension, the number of qubits, and the variational circuit depth, influence the classification performance.

1) ISOLATING THE EFFECT OF THE FEATURE COMPRESSION LAYER

To examine the effect of the downstream classifier under a shared compressed feature space, we evaluate multiple classifier heads or alternative final classification layers operating on the same frozen four-dimensional representation produced by the TF-IDF compressor. Table 5 summarizes the results obtained by a linear classifier head (4→2), linear

¹The source code and implementation details are available at https://github.com/AnnyshaHuzzat/Primer_QNN_Text_Classification.

TABLE 5. Ablation study isolating the effect of the TF-IDF compression layer. Unlike Table 3, where the 5000 → 4 compression stage is learned jointly during model training, here the linear compressor (5000 → 4) is first trained and then frozen, producing a fixed four-dimensional representation shared across all models. Different classifier heads, including a linear classifier head (4 → 2), a linear SVM and the proposed QNN head, are evaluated on the same features to assess the contribution of the quantum circuit independent of the classical compression stage.

Dataset	Classifier Head	Accuracy	F1-score	AUC-ROC
MR	Linear Classifier Head (4 → 2)	0.7760	0.7724	0.8700
	Linear SVM	0.7873	0.7872	0.8700
	QNN Head	0.7901	0.7898	0.8701
CR	Linear Classifier Head (4 → 2)	0.8073	0.7981	0.8937
	Linear SVM	0.8255	0.8239	0.8887
	QNN Head	0.8099	0.8048	0.8932
SUBJ	Linear Classifier Head (4 → 2)	0.9100	0.9100	0.9675
	Linear SVM	0.9050	0.9050	0.9673
	QNN Head	0.9120	0.9120	0.9674
MPQA	Linear Classifier Head (4 → 2)	0.8797	0.8768	0.9227
	Linear SVM	0.8647	0.8635	0.9174
	QNN Head	0.8806	0.8775	0.9224
SST	Linear Classifier Head (4 → 2)	0.8561	0.8557	0.9338
	Linear SVM	0.8567	0.8562	0.9338
	QNN Head	0.8567	0.8563	0.9338

TABLE 6. Ablation study on the MR dataset evaluating the impact of maximum TF-IDF feature dimension, number of qubits, and number of variational layers on the model performance. All experiments are conducted using a fixed learning rate of 0.001 and a batch size of 32.

Maximum Feature Dimension	Number of Qubits	Number of Variational Layers	Accuracy	F1-score	AUC-ROC
500	4	2	0.7338	0.7311	0.8022
1000	4	2	0.7535	0.7544	0.8321
1500	4	2	0.7619	0.7613	0.8410
2000	4	2	0.7657	0.7655	0.8459
2500	4	2	0.7657	0.7628	0.8472
3000	4	2	0.7601	0.7660	0.8460
4000	4	2	0.7732	0.7788	0.8570
5000	4	2	0.7844	0.7843	0.8659
5000	5	4	0.7769	0.7759	0.8548
5000	5	3	0.7807	0.7841	0.8568
5000	5	2	0.7741	0.7791	0.8547
5000	4	3	0.7788	0.7748	0.8586

TABLE 7. Ablation study results across multiple datasets showing the effect of the number of qubits and variational layer on performance metrics. All experiments were conducted using a fixed feature dimension of 5000, learning rate of 0.001 and a batch size of 32.

Dataset	Number of Qubits	Number of Variational Layers	Accuracy	F1-score	AUC-ROC
CR	4	2	0.8307	0.8292	0.8952
	4	3	0.8307	0.8301	0.8945
	5	2	0.8203	0.8176	0.8941
	5	3	0.8229	0.8229	0.8947
	5	4	0.8255	0.8251	0.8925
SUBJ	4	2	0.9170	0.9130	0.9690
	5	2	0.9050	0.9050	0.9681
	4	3	0.9140	0.9140	0.9682
MPQA	4	2	0.8731	0.8706	0.9271
	4	3	0.8722	0.8695	0.9271
	5	2	0.8675	0.8664	0.9176

support vector machine (SVM), and the proposed QNN head operating on this identical feature space. Overall, the results indicate that the frozen compressed representation is already highly informative, as all classifier heads achieve a broadly comparable performance across datasets. At the same time, the downstream classifier still influences the final outcome, and the QNN head remains competitive with the classical alternatives.

For example, on the MR dataset, the QNN head achieves the strongest performance among the compared models,

slightly outperforming both linear SVM and the linear classifier head across all evaluation metrics. On the CR dataset, linear SVM achieves the highest accuracy and F1-score, while the QNN head remains competitive with similar AUC-ROC values. On the SUBJ dataset, the QNN head achieves the highest accuracy and F1-score, with an AUC-ROC nearly identical to the classical models. On MPQA, the QNN head again provides the strongest accuracy and F1-score, whereas the linear classifier head obtains a slightly higher AUC-ROC than that obtained by the QNN head.

TABLE 8. Comparison of CNOT and IsingZZ entanglement for a four-qubit, two-layer quantum circuit with 5000 TF-IDF features and a learning rate of 0.001 across multiple datasets.

Dataset	Entanglement	Accuracy	F1-score	AUC-ROC
MR	CNOT (proposed)	0.7844	0.7843	0.8659
	IsingZZ	0.7760	0.7758	0.8575
CR	CNOT (proposed)	0.8307	0.8292	0.8952
	IsingZZ	0.8125	0.8073	0.8936
SUBJ	CNOT (proposed)	0.9170	0.9130	0.9690
	IsingZZ	0.9110	0.9110	0.9680
MPQA	CNOT (proposed)	0.8731	0.8706	0.9271
	IsingZZ	0.8647	0.8633	0.9195

Finally, on the SST dataset, all classifier heads produce nearly identical results, indicating that the compressed representation itself captures most of the discriminative information for this task.

These results suggest that the classical compression stage performs a substantial portion of the representation learning, which is expected given the high dimensionality of the original TF-IDF features. However, the choice of classifier head still affects the final performance. Under the same frozen compressed representation, the QNN head remains competitive with the classical alternatives across datasets, suggesting that the downstream classifier remains a meaningful source of improvement even after the compression stage has been fixed.

2) OPTIMAL NUMBER OF MAXIMUM TF-IDF FEATURE DIMENSIONS

The primary objective of our study is to investigate how quantum circuits with a minimal number of qubits and variational layers can effectively perform text classification when combined with a simple classical feature extraction technique, such as TF-IDF. As shown in Table 6, we begin by experimenting with different feature dimensions using the MR dataset. Specifically, we test with maximum feature sizes of 500, 1000, 1500, 2000, and higher. The results demonstrate that increasing the feature dimension improves the performance up to a certain point, with the best results achieved when the maximum feature size is set to 5000. This setting is therefore adopted for all subsequent experiments.

3) OPTIMAL NUMBER OF QUBITS AND VARIATIONAL LAYERS

We systematically vary the number of qubits and variational layers in the quantum circuit to evaluate their impact on model performance. Due to the substantially larger size of the SST dataset and the computational cost of repeated quantum simulations, the extended ablations on qubit count, circuit depth, and circuit design were limited to the MR, CR, SUBJ and MPQA datasets. The results of these experiments using the MR dataset are also summarized in Table 6, which highlights the trade-off between circuit complexity and classification accuracy. The results in Table 7 further indicate that even with a compact quantum configuration using four qubits and a variational depth of two, our model consistently achieves the best performance across the remaining datasets.

4) JUSTIFICATION FOR THE CHOSEN QUANTUM CIRCUIT CONFIGURATION

The proposed quantum-classical architecture employs three parameterized single-qubit rotation gates (R_X , R_Y , R_Z) followed by a linear chain of CNOT entanglement operations as shown in Fig. 5. This design corresponds to a compact VQC with four qubits and two layers, as defined in the implementation. To validate this configuration, we conduct two sets of controlled experiments focusing on (i) the choice of entangling gates, and (ii) the effect of introducing the Hadamard gates prior to data embedding.

Table 8 compares CNOT-based entanglement with IsingZZ entanglement across five benchmark datasets while keeping all other architectural and training parameters fixed. The results show that CNOT gives better accuracy, F1-score and AUC-ROC across all datasets, with more stable performance overall, especially on the MR and CR datasets. This suggests that CNOT-based entanglement is more suitable for our setting. These findings are also consistent with earlier studies reporting that CNOT-based variational circuits can provide strong expressibility and stable training in shallow VQCs, while Ising-type interactions may be more sensitive to circuit depth and parameter initialization [50], [51].

To further investigate whether increasing the quantum expressivity could enhance the model performance, we introduce a Hadamard gate prior to Angle Embedding so that the circuit begins in a uniform superposition state, where each qubit is initialized with an equal probability amplitude across all computational basis states. As shown in Table 9, this modification does not lead to consistent improvements. In several cases, it produces a slight degradation. The effect is particularly evident on the MR and SUBJ datasets, where the Hadamard initialization results in a lower accuracy and F1-scores compared to the proposed circuit that starts from the computational $|0\rangle$ state.

These trends suggest that injecting superposition at the initial stage can introduce an additional variance or noise in the encoded feature space [52], especially when the TF-IDF vectors already distribute the information across thousands of dimensions. This can reduce the performance stability, as the model must disentangle additional quantum variability on top of the classical feature dispersion. Based on these observations, we conclude that initializing all qubits in the $|0\rangle$ state provides a more reliable and controlled starting point for our proposed text classification model.

B. MODEL LIMITATIONS AND FUTURE WORK

In this study, we intentionally adopted a simple and interpretable feature extraction pipeline to isolate and analyze the behavior of the QNN component and to provide an accessible entry point for researchers new to QML. However, we would like to emphasize that our proposed framework is fully reproducible and can readily accommodate more advanced text representations in lieu of TF-IDF. In modern NLP tasks, the TF-IDF suffers from limitations because it represents text as an unordered bag-of-words representation and, therefore, discards word order, contextual dependencies, and semantic

TABLE 9. Effect of introducing a Hadamard (H) gate prior to Angle Embedding compared to circuits using only rotational gates. Results are reported for both CNOT and IsingZZ entanglement, with all experiments using four-qubits and two variational layers.

Dataset	Gate Configuration	Accuracy	F1-score	AUC-ROC
MR	$R_X, R_Y, R_Z + \text{CNOT}$ (proposed)	0.7807	0.7780	0.8628
	$H + R_X, R_Y, R_Z + \text{CNOT}$	0.7723	0.7722	0.8528
	$H + R_X, R_Y, R_Z + \text{IsingZZ}$	0.7713	0.7711	0.8517
CR	$R_X, R_Y, R_Z + \text{CNOT}$ (proposed)	0.8307	0.8292	0.8952
	$H + R_X, R_Y, R_Z + \text{CNOT}$	0.8281	0.8257	0.8958
	$H + R_X, R_Y, R_Z + \text{IsingZZ}$	0.8255	0.8236	0.8953
SUBJ	$R_X, R_Y, R_Z + \text{CNOT}$ (proposed)	0.9170	0.9130	0.9690
	$H + R_X, R_Y, R_Z + \text{CNOT}$	0.9120	0.9120	0.9685
	$H + R_X, R_Y, R_Z + \text{IsingZZ}$	0.9100	0.9100	0.9680
MPQA	$R_X, R_Y, R_Z + \text{CNOT}$ (proposed)	0.8731	0.8706	0.9271
	$H + R_X, R_Y, R_Z + \text{CNOT}$	0.8703	0.8677	0.9178
	$H + R_X, R_Y, R_Z + \text{IsingZZ}$	0.8638	0.8625	0.9166

relationships between tokens. As a result, it cannot capture the sequential structure or contextual meaning that are typically exploited by modern language models. In this work, TF-IDF is used intentionally to keep the classical preprocessing stage simple so that the contribution of the quantum classifier can be studied in isolation.

Future extensions of this framework could incorporate richer semantic representations such as Word2Vec [53] and GloVe embeddings [54], as well as contextual token embeddings derived from compact transformer models [55], [56] or other pretrained language models [57], [58]. Such embeddings preserve semantic relationships between words and encode contextual information across tokens, providing a more informative representation of textual data. These embeddings could be projected to a compact dimension and subsequently encoded into quantum states using standard techniques, such as angle encoding or amplitude encoding. This would allow the quantum circuit to operate on semantically meaningful representations while maintaining a small number of qubits.

The current VQC architecture is also intentionally constrained to a small number of qubits and shallow circuit depth in order to remain compatible with widely available quantum simulators and near-term quantum devices. While this design improves training stability and reproducibility, it limits the expressivity of the circuit. In addition, all experiments are conducted using classical simulation of quantum circuits implemented via PennyLane, which provides an idealized execution environment and does not fully capture hardware effects such as noise, finite-shot sampling, and gate imperfections. Consequently, the reported results should be interpreted as performance under noiseless simulation. Finally, the current evaluation focuses on binary text classification tasks. Future work will therefore explore richer text representations, alternative quantum data encoding strategies, more expressive yet trainable circuit architectures, evaluation on noisy quantum hardware, and extension of the framework to broader NLP settings such as multi-class and multi-label classification.

VI. CONCLUSION

In this paper, we presented a compact QNN model for binary text classification that integrated a TF-IDF front-end with a shallow VQC backend, implemented using PennyLane

and trained end-to-end in PyTorch. By compressing high-dimensional TF-IDF features to a four-qubit representation and replacing a classical hidden layer with a two-layer VQC, the proposed model formed a low-dimensional classification pipeline that maintained competitive predictive performance. Experiments on five benchmark datasets showed that the QNN achieved a performance comparable to that of the larger and compact classical NN baselines, while delivering the strongest overall results on the CR, SUBJ and MPQA datasets. Further ablation studies analyzing the compression stage, qubit count, and circuit depth indicated that the classical compression layer produced a compact feature representation that preserved useful information from the original TF-IDF inputs, while the downstream classifier head, including the QNN, determined how these features were mapped to the final class predictions. These findings suggest that shallow, trainable VQCs can serve as competitive classifier components in compact text classification pipelines and motivate future work on richer text representations, more complex NLP tasks, and evaluation on noisy quantum hardware backends.

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